

Creating File and Directory using File class

The **File** class in Java allows you to work with files and directories on your computer's storage. It provides methods to create, check, and manipulate files and folders.

👉 Creating a File Reference

- We need to create a File object that refers to a file location on our hard disk.

```
import java.io.File;

public class LaunchI02 {
    public static void main(String[] args) {

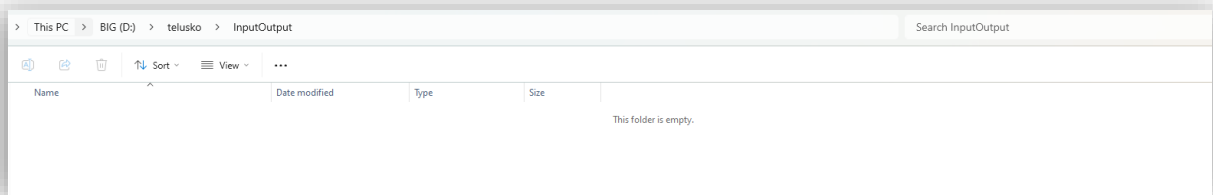
        // Create a File object that refers to a file path
        // Note: This doesn't create the actual file yet, just a reference to where it would be
        File file1 = new File("D:\\telusko\\InputOutput\\java.txt");

        // Check if the file already exists at the specified location
        System.out.println(file1.exists());
    }
}
```

Output:

```
false
```

The output **false** means that the file doesn't exist at the specified location yet. We've only created a reference to a potential file location.



👉 Creating a New File

- We have a file reference, let's actually create the file:

```
import java.io.File;

public class LaunchI02 {
    public static void main(String[] args) {
        try {
            // Create a File object reference
            File file1 = new File("D:\\telusko\\InputOutput\\java.txt");

            // Check if the file exists before trying to create it
            System.out.println("File exists? " + file1.exists());

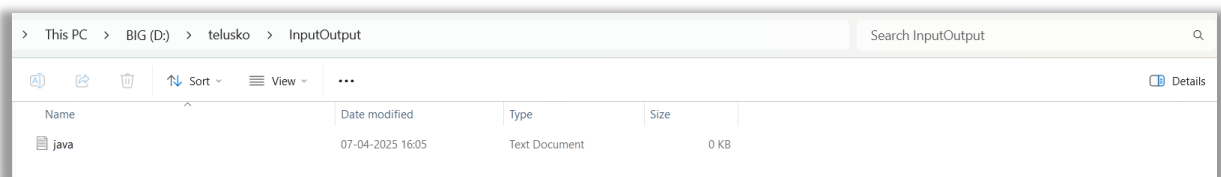
            // Create the actual file on disk
            // createNewFile() returns true if the file was created,
            // and false if it already exists
            System.out.println("File created? " + file1.createNewFile());

        } catch (Exception e) {
            // Handle any errors that might occur
            System.out.println("Some problem occurred: " + e.getMessage());
        }
    }
}
```

Output:

```
File exists? false
File created? true
```

- The createNewFile() method returns true if it successfully creates a new file
- It returns false if the file already exists
- This method can throw exceptions, so we need to use try-catch
- After running this code, you can check your folder and you'll see the actual java.txt file has been created



👉 Creating a Directory

- We can also create directories (folders) using the **File** class:

```
import java.io.File;

public class LaunchIO2 {
    public static void main(String[] args) {
        try {
            // Commented out previous file creation code
            // File file1 = new File("D:\\telusko\\InputOutput\\java.txt");
            // System.out.println("File exists? " + file1.exists());
            // System.out.println("File created? " + file1.createNewFile());

            // Create a File object that refers to a directory path
            File dir = new File("D:\\telusko\\InputOutput\\Directory");

            // Check if the directory already exists
            System.out.println("Directory exists? " + dir.exists());

            // Create the directory if it doesn't exist
            if (!dir.exists()) {
                System.out.println("Directory created? " + dir.mkdir());
            }

        } catch (Exception e) {
            System.out.println("Some problem occurred: " + e.getMessage());
        }
    }
}
```

Output:

```
Directory exists? false
Directory created? true
```

This PC > BIG (D:) > telusko > InputOutput					Search InputOutput		Q
Sort View ...					Details		
Name	Date modified	Type	Size				
java	07-04-2025 16:05	Text Document	0 KB				
Directory	07-04-2025 16:12	File folder					

👉 Key Points to Remember

- The `File` class is in the `java.io` package
- Creating a `File` object doesn't create the actual file/directory
- Use `createNewFile()` to create a file
- Use `mkdir()` or `mkdirs()` to create directories
- Always use try-catch blocks when working with file operations
- Use `exists()` to check if a file or directory already exists
- If a file already exists, `createNewFile()` will return false
- File paths in Java use double backslashes (`\\`) for Windows paths